Reg No.

SAITHABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT EXAM-2

Programme: B.E CSE SP,ECE,ECE DS,EEE,MECH,MECHATRO,CIVIL,AERO Max. Marks: 50
Course Code: SCYB1101 Time: 2 Hours
Course Name: CHEMISTRY Sem: II
Batch : 2025-2029 SET-A Date: 17.2.2026

Part-A Answer ALL the questions (7 × 2=14)

Q.No	Questions	CO
1.	Calculate the zero point energy for a particle of mass 9.1×10^{-31} Kg moving in 1D box of length 15Å. ($h=6.626 \times 10^{-34}$ Js)	1
2.	Classify solids based on Energy band gap.	1
3.	Differentiate n-type and p-type semiconductors.	1
4.	Identify which of the following is allowed and not allowed. a) $n=0, l=2, m_l=0$ and b) $n=2, l=1, m_l=0$	1
5.	State the conditions for vibrational transition to occur.	2
6.	Which of the following are microwave active molecule a) CS_2, b) HCl, c) N_2, d) O_2	2
7.	Identify the molecule based on moment of Inertia a) $I_A=I_B=I_C$ and b) $I_C=I_B \neq I_A$ and $I_A = 0$	2

Part-B Answer ALL the questions (3 × 12 = 36)

8.	Formulate the wave function and Energy for a particle in 1D box.	1
	(OR)	
9.	Explain the significance of Quantum numbers.	1
10.	Discuss the shapes of various atomic orbitals using the boundary surface diagram.	1
	(OR)	
11.	With neat sketch explain the vibrational modes for i) CO_2 molecule and ii) H_2O molecule	2
12.	Explain the rotational spectra for a diatomic molecule and obtain the expression for first three rotational transitions.	2
	(OR)	
13.	Discuss the various possible electronic transitions in organic molecules.	2